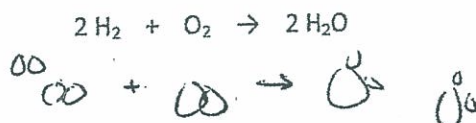


SOLUTIONS

CHAPTER 9.5-9.6

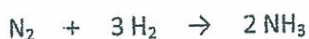
Stoichiometry

In a balanced chemical equation, the coefficients (balancing numbers), show the ratio in which the reactant species combine to form products. For example, on the smallest scale the equation below means that 2 molecules of hydrogen will react with 1 molecule of oxygen to produce 2 molecules of water.



However, reactions don't usually happen on a scale that small. A more realistic way to interpret the coefficients is to say that 2 moles of hydrogen will react with 1 mole of oxygen to produce 2 moles of water.

Look at the following equation and complete the table below;



Moles of N ₂	Moles of H ₂	Moles of NH ₃
2 mol	6	4
4	12 mol	8
10	30	20 mol
0.1 mol	0.3	0.2
1/3	1 mol	2/3

This is simple if you are using whole / simple numbers, but this is rarely the case.

How many moles of hydrogen would be needed to react with 0.559 mol of nitrogen?

$$n(\text{H}_2) = \frac{0.559}{1} \times 3 = 1.677 \text{ mol}$$

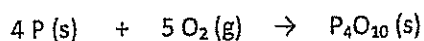
How many moles of nitrogen would be consumed in the production of 1.784 mol of ammonia?

$$n(\text{N}_2) = \frac{1.784}{2} \times 1 = 0.892 \text{ mol}$$

How many moles of NH₃ would be produced if 0.0733 mol of hydrogen was consumed?

$$n(\text{NH}_3) = \frac{0.0733}{3} \times 2 = 0.0489 \text{ mol}$$

Look at the following equation and complete the table below;



Moles of P	Moles of O ₂	Moles of P ₄ O ₁₀
3.54 mol	$n(\text{O}_2) = \frac{3.54}{4} \times 5$ $= 4.425$	$n(\text{P}_4\text{O}_{10}) = \frac{3.54}{4} \times 1$ $= 0.885$
$n(\text{P}) = \frac{0.119}{1} \times 4$ $= 0.476$	$n(\text{O}_2) = \frac{0.119}{1} \times 5$ $= 0.595$	0.119 mol
$n(\text{P}) = \frac{0.863}{5} \times 4$ $= 0.6904$	0.863 mol	$n(\text{P}_4\text{O}_{10}) = \frac{0.863}{5} \times 1$ $= 0.1726$
0.00929	$n(\text{O}_2) = \frac{0.00929}{4} \times 5$ $= 0.0116$	$n(\text{P}_4\text{O}_{10}) = \frac{0.00929}{4} \times 1$ $= 0.00232$
$n(\text{P}) = \frac{2.27}{5} \times 4$ $= 1.816$	2.27 mol	$n(\text{P}_4\text{O}_{10}) = \frac{2.27}{5} \times 1$ $= 0.454$

Look at the following equation and complete the questions below;



- a) How many moles of carbon dioxide would be produced if 7.022 mol of oxygen is consumed?

$$n(\text{CO}_2) = \frac{7.022}{13} \times 8 = 4.321 \text{ mol}$$

- b) How many moles of water would be produced in the combustion of 0.338 mol of butane?

$$n(\text{H}_2\text{O}) = \frac{0.338}{2} \times 10 = 1.69 \text{ mol}$$

- c) How many moles of butane would be consumed if 3.901 mol of carbon dioxide was produced?

$$n(\text{C}_4\text{H}_{10}) = \frac{3.901}{8} \times 2 = 0.975 \text{ mol}$$

- d) How many moles of oxygen would be required to produce 0.0828 mol of water?

$$n(\text{O}_2) = \frac{0.0828}{10} \times 13 = 0.108 \text{ mol}$$

e.g. $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$
21.9g ?g.

$$n(\text{HCl}) = \frac{0.2188}{1} \times 2 = 0.4376 \text{ mol}$$

$$2 \text{ Al (s)} + 3 \text{ Cl}_2 \text{ (g)} \rightarrow 2 \text{ AlCl}_3 \text{ (s)}$$

675g ?g 4.99g

- $$n(\text{Al}) = \frac{m}{M} = \frac{6.75}{26.98} = 0.25019 \text{ mol.}$$

$$m(\text{AlCl}_3) = nM = 0.25019 \times (26.98 + 3 \times 35.45) = 33.4 \text{ g}$$

- $$n(\text{AlCl}_3) = \frac{m}{M} = \frac{14.9}{26.98 + 3 \times 35.45} = 0.11175 \text{ mol}$$

$$n(\text{Cl}_2) = \frac{0.1175}{2} \times 3 = 0.1676 \text{ mol}$$

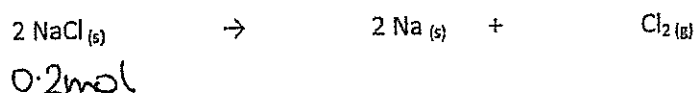
$$m_{CCl_2} = m_M = 0.1676 \times (35.45 \times 2) = 11.9 \text{ g}$$

PRACTICE CALCULATIONS

1. If 0.2 mole of sodium chloride is electrolysed, determine;

a) the number of moles of sodium produced

b) the number of moles of chlorine produced.



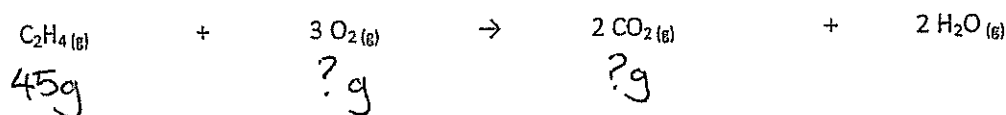
$$n(\text{Na}) = 0.2 \text{ mol}$$

$$n(\text{Cl}_2) = 0.1 \text{ mol}$$

2. If you start with 45 g of ethylene (C_2H_4), determine;

a) the mass of oxygen that is required to react

b) the mass of carbon dioxide that will be produced



$$a) \quad n(\text{C}_2\text{H}_4) = \frac{m}{M} = \frac{45}{2 \times 12.01 + 4 \times 1.008} = 1.6042 \text{ mol}$$

$$n(\text{O}_2) = \frac{1.6042}{1} \times 3 = 4.8125 \text{ mol}$$

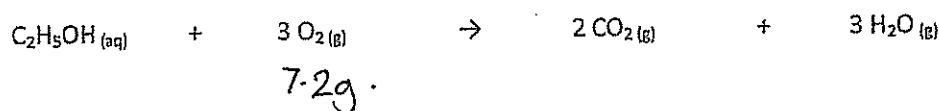
$$m(\text{O}_2) = nM = 4.8125 \times 32 = 154 \text{ g}$$

$$b) \quad n(\text{CO}_2) = \frac{1.6042}{1} \times 2 = 3.2084 \text{ mol}$$

$$m(\text{CO}_2) = nM = 3.2084 \times (44.01) = 141 \text{ g}$$

3. Ethanol burns according to the equation below. If 7.2 g of oxygen was used up, determine;

- a) the number of moles of ethanol consumed
b) the mass of carbon dioxide produced



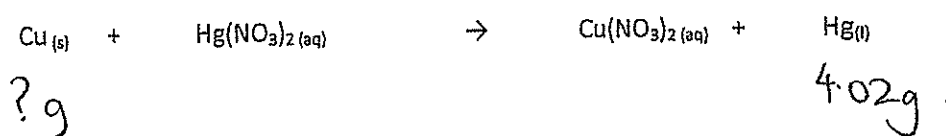
$$a) \cdot n(O_2) = \frac{m}{M} = \frac{7.2}{32} = 0.225 \text{ mol}$$

$$n(\text{C}_2\text{H}_5\text{OH}) = \frac{0.225}{3} \times 1 = 0.075 \text{ mol}$$

$$b) n(\text{CO}_2) = \frac{0.225}{3} \times 2 = 0.15 \text{ mol}$$

$$m(\text{CO}_2) = nM = 0.15 \times 44.01 = 6.60 \text{ g}$$

4. When copper is placed in a solution of mercury (II) nitrate, mercury is displaced and the copper is completely dissolved. If 4.02 g of mercury is produced, determine the mass of copper metal added.

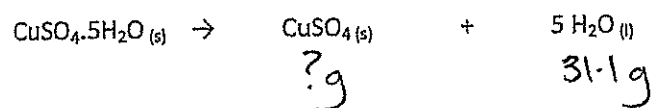


$$n(\text{Hg}) = \frac{m}{M} = \frac{4.02}{200.6} = 0.02004 \text{ mol}$$

$$n(w) = 0.02004 \text{ mol}$$

$$m_{\text{ant.}} = nM = 0.02004 \times 63.55 = 1.27 \text{ g}$$

5. When copper (II) sulfate pentahydrate is heated, the water can be driven off. If 31.1 g of water is formed, what mass of copper sulfate is also formed?

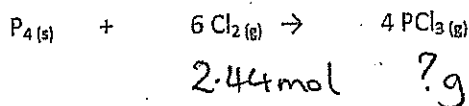


$$n(\text{H}_2\text{O}) = \frac{m}{M} = \frac{31.1}{18.016} = 1.7262 \text{ mol}$$

$$n(\text{CuSO}_4) = \frac{1.7262}{5} \times 1 = 0.34525 \text{ mol}$$

$$m(\text{CuSO}_4) = nM = 0.34525 \times (63.55 + 32.06 + 4 \times 16) \\ = 55.1 \text{ g}$$

6. If 2.44 mole of chlorine is consumed, what mass of phosphorus trichloride is formed?



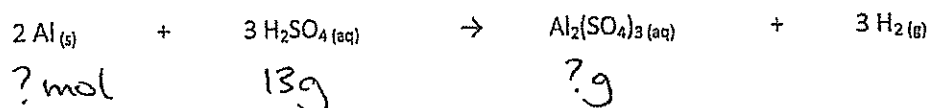
$$n(\text{PCl}_3) = \frac{2.44}{6} \times 4 = 1.6267 \text{ mol}$$

$$m(\text{PCl}_3) = nM = 1.6267 \times (30.97 + 3 \times 35.45) \\ = 223 \text{ g}$$

7. If 13 g of sulfuric acid was consumed in the reaction, determine;

a) the number of moles of aluminium reacted

b) the mass of aluminium sulfate produced



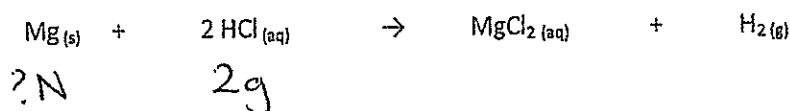
$$a) n(\text{H}_2\text{SO}_4) = \frac{m}{M} = \frac{13}{2 \times 1.008 + 32.06 + 4 \times 16} = 0.13255 \text{ mol}$$

$$n(\text{Al}) = \frac{0.13255}{3} \times 2 = 0.0884 \text{ mol}$$

$$b) n(\text{Al}_2(\text{SO}_4)_3) = \frac{0.13255}{3} \times 1 = 0.04418 \text{ mol}$$

$$\begin{aligned}
 m(\text{Al}_2(\text{SO}_4)_3) &= nM = 0.04418 \times (2 \times 26.98 + 3 \times 32.06 + 12 \times 16) \\
 &= 15.1 \text{ g}
 \end{aligned}$$

8. If 2 g of hydrochloric acid was consumed in the reaction, calculate how many atoms of magnesium were used.

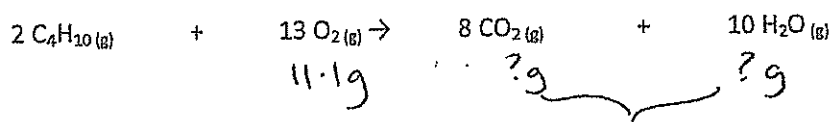


$$n(\text{HCl}) = \frac{m}{M} = \frac{2}{1.008 + 35.45} = 0.054858 \text{ mol}$$

$$n(\text{Mg}) = \frac{0.054858}{2} \times 1 = 0.02743 \text{ mol}$$

$$\begin{aligned}
 N(\text{Mg}) &= nN_A = 0.02743 \times 6.022 \times 10^{23} \\
 &= 1.65 \times 10^{22} \text{ atoms}
 \end{aligned}$$

9. When butane burns in air, carbon dioxide and water vapour are formed. If 11.1 g of oxygen was consumed in the reaction, calculate the total mass of gaseous products formed.



$$n(\text{O}_2) = \frac{m}{M} = \frac{11.1}{32} = 0.34688 \text{ mol}$$

$$n(\text{CO}_2) = \frac{0.34688}{13} \times 8 = 0.213465 \text{ mol}$$

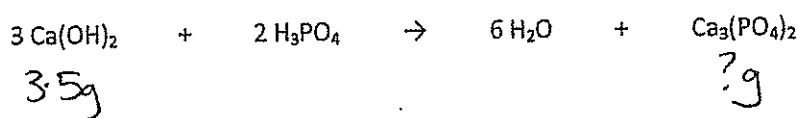
$$m(\text{CO}_2) = nM = 0.213465 \times 44.01 = 9.3946 \text{ g}$$

$$n(\text{H}_2\text{O}) = \frac{0.34688}{13} \times 10 = 0.266831 \text{ mol}$$

$$m(\text{H}_2\text{O}) = nM = 0.266831 \times 18.016 = 4.8072 \text{ g}$$

$$m(\text{total gases}) = 14.2 \text{ g}$$

10. Calculate the mass of calcium phosphate produced when 3.5 g of calcium hydroxide reacts with phosphoric acid (H_3PO_4).



$$n(\text{Ca}(\text{OH})_2) = \frac{m}{M} = \frac{3.5}{40.08 + 2 \times 16 + 2 \times 1.008} = 0.04724 \text{ mol}$$

$$n(\text{Ca}_3(\text{PO}_4)_2) = \frac{0.04724}{3} \times 1 = 0.015745 \text{ mol}$$

$$m(\text{Ca}_3(\text{PO}_4)_2) = nM = 0.015745 \times (3 \times 40.08 + 2 \times 30.97 + 8 \times 16) = 4.88 \text{ g}$$

11. If 4.3 g of ethane (C_2H_6) is burnt according to the reaction below, calculate;

a) the number of moles of oxygen consumed

b) the mass of water produced



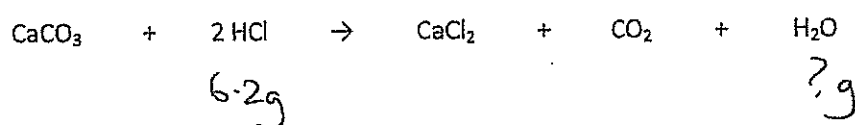
$$a) n(C_2H_6) = \frac{m}{M} = \frac{4.3}{2 \times 12.01 + 6 \times 1.008} = 0.14301 \text{ mol}$$

$$n(O_2) = \frac{0.14301}{2} \times 7 = 0.5005 \text{ mol}$$

$$b) n(H_2O) = \frac{0.14301}{2} \times 6 = 0.42903 \text{ mol}$$

$$\begin{aligned}
 m(H_2O) &= nM = 0.42903 \times (18.016) \\
 &= 7.73g
 \end{aligned}$$

12. What mass of water is produced when 6.2 g of hydrochloric acid reacts with calcium carbonate.

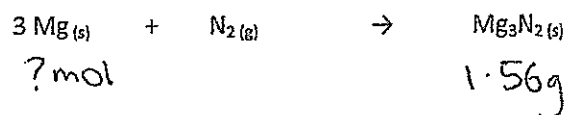


$$n(HCl) = \frac{m}{M} = \frac{6.2}{1.008 + 35.45} = 0.17006 \text{ mol}$$

$$n(H_2O) = \frac{0.17006}{2} \times 1 = 0.085029 \text{ mol}$$

$$\begin{aligned}
 m(H_2O) &= 0.085029 \times 18.016 \\
 &= 1.53g
 \end{aligned}$$

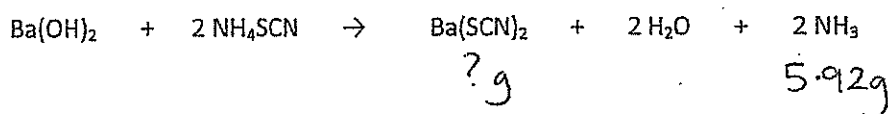
13. Calculate the number of moles of magnesium metal required to form 1.56 g of magnesium nitride by the following reaction.



$$n(\text{Mg}_3\text{N}_2) = \frac{m}{M} = \frac{1.56}{3 \times 24.31 + 2 \times 14.01} = 0.015453 \text{ mol}$$

$$n(\text{Mg}) = \frac{0.015453}{1} \times 3 = 0.0464 \text{ mol}$$

14. Calculate the mass of barium thiocyanate produced if 5.92 g of ammonia was also produced.



$$n(\text{NH}_3) = \frac{m}{M} = \frac{5.92}{14.01 + 3 \times 1.008} = 0.34754 \text{ mol}$$

$$n(\text{Ba(SCN)}_2) = \frac{0.34754}{2} \times 1 = 0.1738 \text{ mol}$$

$$\begin{aligned}
 m(\text{Ba(SCN)}_2) &= nM = 0.1738 \times (137.3 + 2 \times 32.06 + 2 \times 12.01 + 2 \times 14.01) \\
 &= 44.0 \text{ g}
 \end{aligned}$$